

Chapter 14 - SN76489

The SN76489 is the Texas Instruments three-voice tone generator used in the BBC Micro, the Sega Master System, and (with one register-format change) the Sega Game Gear. It produces three square-wave channels and one noise channel, with 4-bit volume attenuation per voice. Intuition Engine's SN76489 is faithful to the original chip.

14.1 What the SN76489 can show

Item	Value
Tone channels	3
Noise channel	1, white or periodic
Volume control	Per-channel 4-bit attenuation
Register access	Single byte-stream write port

14.2 The register block

The SN76489 sits in a tiny three-byte block at 0xF0C30–0xF0C3F:

Address	Name	Purpose
0xF0C30	SN_PORT_WRITE	Byte-stream write port.
0xF0C31	SN_PORT_READY	Bit 0 = ready for another byte (read-only).
0xF0C32	SN_PORT_MODE	0 = 15-bit LFSR, 1 = 16-bit LFSR.

Programming is by **byte stream**. Each byte writes one of two kinds of message into the chip:

Bit 7	Message kind
1	Latch + data low: bits 4–6 select register, bits 0–3 are the four low data bits.
0	Data high: bits 0–5 are the next six bits of the most recently latched register.

The register field is three bits: bits 4–5 choose the channel (0–3), bit 6 chooses between frequency (0) and attenuation (1). Channels 0–2 are tone; channel 3 is noise.

So a 10-bit frequency value is written as **two** bytes: first a latch byte with the low 4 bits, then a data byte with the next 6 bits. A 4-bit attenuation needs only one latch byte.

14.3 Noise generator

The noise channel uses a linear-feedback shift register; the SN_PORT_MODE register selects whether it is the 15-bit form (SN76489_MODE_LFSR_15) used by the SMS or the 16-bit form (SN76489_MODE_LFSR_16) used by the Game Gear. The frequency field of the noise channel chooses one of three fixed divider rates or "use channel 2's divider", so the noise pitch can be locked to a melody.

14.4 Clocks

Three clock constants ship for fidelity:

Constant	Value (Hz)
SN_CLOCK_NTSC	3,579,545
SN_CLOCK_PAL	3,546,893
SN_CLOCK_BBC	4,000,000

The output frequency of a tone channel is $\text{clock} / (32 \times \text{divider})$.

14.5 BASIC and SoundChip relationship

The BASIC SOUND keyword does **not** target the SN76489 directly; for direct access, write bytes through SN_PORT_WRITE using POKE. The cleanest way to use SN-style noise from BASIC is the SoundChip's noise mode (Chapter 12) with NOISEMODE set to one of:

NOISEMODE	Algorithm
5	SN76489 15-bit white noise
6	SN76489 15-bit periodic noise
7	SN76489 16-bit white noise
8	SN76489 16-bit periodic noise

14.6 A short example

A BASIC fragment that plays a 440 Hz tone on SN76489 channel 0 at full volume:

```
10 POKE &H000F0800, 1          : REM AUDIO_CTRL = on
20 REM divider for 440 Hz at NTSC clock = 3579545 / (32 * 440) ≈ 254
30 POKE &H000F0C30, &H80 OR 254 AND 15 : REM latch byte: ch 0, freq, low 4 bits
40 POKE &H000F0C30, (254 >> 4) AND 63   : REM data byte: top 6 bits
50 POKE &H000F0C30, &H90                : REM latch: ch 0, attenuation, value 0 (loudest)
```

14.7 Putting it together

The SN76489 is the smallest and simplest of the heritage chips. Most programs use it through the byte-stream port for direct access, or through the SoundChip's emulated noise modes. The next chapter covers the SID family, the most musically capable of the heritage engines.